Dyskeratosis Congenita, Autosomal Recessive 8 (DKCB8): A Comprehensive Disease Characterization

Disease: Dyskeratosis Congenita Autosomal Recessive 8 (DKCB8) **MONDO ID:** MONDO:0859319 · **OMIM:** 620133 · **Category:** Mendelian (telomere biology disorder) **Causal locus:** *TYMS–ENOSF1* (digenic), chromosome 18p11.32

Summary

Dyskeratosis Congenita, Autosomal Recessive 8 (DKCB8) is an ultra-rare, childhood-onset inherited bone marrow failure and telomere biology disorder (TBD) defined by a distinctive **digenic** genetic architecture at the *TYMS–ENOSF1* locus. Unlike classic single-gene recessive dyskeratosis congenita, DKCB8 arises when an individual inherits a **loss-of-function coding variant in *TYMS*** (thymidylate synthase) from one parent and a specific **haplotype with rare variants in the antisense regulator *ENOSF1*** (enolase superfamily 1) from the other parent. Because *ENOSF1* post-transcriptionally silences the remaining wild-type *TYMS* allele, the net effect is severe thymidylate synthase deficiency — even though neither parent alone is affected and one parent carries an entirely wild-type *TYMS* coding sequence. This "pseudo-recessive" inheritance was established by Tummala et al. (Blood, 2022) across eight independent DC families and confirmed by an independent 2025 case report [PMID: 35931051; P 40207375].

Mechanistically, thymidylate synthase catalyzes the sole *de novo* route to dTMP (dUMP → dTMP, using 5,10-methylenetetrahydrofolate). Its deficiency in DKCB8 depletes the dTMP/dTTP pool and distorts the balance of cellular deoxyribonucleotides, promoting **uracil misincorporation into DNA, base-excision-repair–mediated strand breaks, replication-fork collapse, and genotoxic stress**, which in turn produce **abnormal telomere maintenance** and stem-cell attrition. The clinical consequence is the classic mucocutaneous triad of dyskeratosis congenita

(reticulate skin pigmentation, nail dystrophy, oral leukoplakia) together with progressive bone marrow failure, pulmonary and hepatic fibrosis, and an elevated risk of myelodysplastic syndrome, leukemia, and squamous cell carcinoma.

Clinically, DKCB8 is diagnosed and managed as part of the broader dyskeratosis congenita / TBD spectrum: very short telomeres (flow-FISH below the 1st percentile) provide the screening biomarker, and targeted sequencing of the *TYMS–ENOSF1* locus (rather than standard single-gene panels) is required to capture the digenic lesion. Management centers on androgens (danazol/oxymetholone) for cytopenias and fludarabine-based reduced-intensity allogeneic hematopoietic stem cell transplantation (HSCT) as the only cure for marrow failure, while non-hematopoietic complications (pulmonary/hepatic fibrosis, malignancy) remain the principal drivers of late mortality. A mechanistically-inferred, DKCB8-specific caution is that patient cells are hypersensitive to fluoropyrimidines (5-fluorouracil, capecitabine) and antifolates, which target thymidylate synthase — these agents should be avoided.

Section 1 — Disease Information

Overview. DKCB8 is a Mendelian, autosomal recessive (digenic) subtype of dyskeratosis congenita, itself a prototypical **telomere biology disorder**. Dyskeratosis congenita is a progressive bone-marrow-failure syndrome classically presenting with the ectodermal/ mucocutaneous triad of **reticulate skin pigmentation, nail dystrophy, and oral leukoplakia**, with a wide spectrum of multisystem complications and cancer predisposition [PMID: 42625322;

P 35097237

].

> "Classically, it presents with the ectodermal triad of reticulate skin pigmentation, nail dystrophy, and oral leukoplakia" — [PMID: 42625322]

Key identifiers. Cross-references retrieved from EBI OLS4 / MONDO (Finding F008):

Resource	Identifier
MONDO	MONDO:0859319 ("dyskeratosis congenita, autosomal recessive 8")
OMIM	620133 (equivalentTo)
GARD	0026695
MedGen	C1824030
UMLS	C5774257
Abbreviation	DKCB8
Parent concept	dyskeratosis congenita (MONDO:0015780; MeSH D019871; Orphanet ORPHA:1775)

The causal locus per OMIM 620133 is *TYMS* (with *ENOSF1*), referencing Tummala et al. 2022 [PMID: 35931051].

Synonyms / alternative names. "DKCB8"; "dyskeratosis congenita, autosomal recessive 8"; within the literature, "TYMS-ENOSF1 dyskeratosis congenita" and "thymidylate synthase deficiency dyskeratosis congenita" [PMID: 40207375].

Source of information. The DKCB8 entry is derived from **aggregated disease-level resources** (OMIM, MONDO) built on a small number of **individual-patient reports** — the original eight-family cohort [PMID: 35931051] and subsequent single case reports [PMID: 40207375]. There is no EHR-scale dataset for this ultra-rare entity.

Section 2 — Etiology

Primary cause — genetic, digenic. DKCB8 is caused by **germline digenic *TYMS–ENOSF1* variants** producing thymidylate synthase deficiency (Finding F001). Tummala et al. identified heterozygous germline *TYMS* variants across eight independent DC families; crucially, in each family **one parent carried a wild-type *TYMS* coding sequence**, while the other transmitted a **specific *ENOSF1* haplotype plus rare *ENOSF1* variants**:

> "we have identified a remarkable series of heterozygous germline variants in the gene encoding thymidylate synthase (TYMS). Although the inheritance appeared to be autosomal recessive, one parent in each family had a wild-type TYMS coding sequence.

> Targeted genomic sequencing identified a specific haplotype and rare variants in the naturally occurring TYMS antisense regulator ENOSF1 (enolase super family 1) inherited from the other parent" — [PMID: 35931051]

An independent 2025 report confirmed the mechanism:

> "compound heterozygosity for loss of function variants in TYMS and a specific haplotype of its antisense regulator ENOSFI (enolase super family 1) causes digenic DC" — [PMID: 40207375]

Genetic risk factors. The obligate risk determinants are (i) a *TYMS* loss-of-function coding allele and (ii) a permissive *ENOSF1* antisense haplotype carrying rare variants in *trans*. Both are required; neither alone is sufficient (Findings F001, F007).

Environmental risk factors. No environmental cause is required for disease. However, a clinically important **gene–environment interaction** exists: because thymidylate synthase is the pharmacologic target of fluoropyrimidines, DKCB8 confers **hypersensitivity to 5-fluorouracil and folate-antagonist chemotherapy** (Finding F009):

> "hypersensitivity to the TYMS-specific inhibitor 5-fluorouracil" — [PMID: 35931051]

TYMS uses 5,10-methylenetetrahydrofolate as cofactor, linking enzyme activity to dietary **folate / one-carbon metabolism** [PMID: 37183313; P 28461497]. This provides a plausible (though not clinically demonstrated for DKCB8) axis of modifiable risk.

Protective factors. No specific protective variants or environmental protective factors have been reported for DKCB8. Avoidance of fluoropyrimidine/antifolate exposure is a mechanistically-inferred protective action (not a natural protective factor).

Gene–environment interaction. The principal, evidence-based GxE relationship is pharmacogenomic (TYMS deficiency × fluoropyrimidine/antifolate exposure), detailed in Sections 5 and 12 [PMID: 35931051; P 27569869; P 22496803].

Section 3 — Phenotypes

The phenotypic spectrum of DKCB8 mirrors classic dyskeratosis congenita (Finding F003). Reported features, with suggested HPO terms:

Phenotype	Type	HPO term (suggested)	Notes / frequency
Reticulate skin hyperpigmentation	Physical manifestation	HP:0007441 (reticulate skin pigmentation)	Core triad; childhood onset
Nail dystrophy	Physical manifestation	HP:0008404 (nail dystrophy)	Core triad
Oral leukoplakia	Clinical sign	HP:0002745 (oral leukoplakia)	Core triad; may progress to SCC
Diffuse hyperpigmentation + punctate hypopigmented macules	Physical manifestation	HP:0007441 / HP:0001010	Documented in confirmed DKCB8 case
Sparse hair	Physical manifestation	HP:0008070 (sparse hair)	Reported in DKCB8 case
Bone marrow failure / cytopenias	Laboratory abnormality	HP:0005528; HP:0001903; HP:0001873	Often first–second decade; drives mortality
Poor growth / failure to thrive	Clinical sign	HP:0001508 (failure to thrive)	Reported in DKCB8 case
Feeding difficulties	Symptom	HP:0011968 (feeding difficulties)	Reported in DKCB8 case
Strabismus	Clinical sign	HP:0000486 (strabismus)	Reported in DKCB8 case
Oral lichenoid lesions	Clinical sign	HP:0030955 / related	Broader DC oral spectrum
Pulmonary fibrosis	Physical manifestation	HP:0002206 (pulmonary fibrosis)	Adult / late; major late-mortality driver
Pulmonary arteriovenous malformations	Physical manifestation	HP:0002638 / related	DC complication
Liver fibrosis	Physical manifestation	HP:0001395 (hepatic fibrosis)	Frequently subclinical

Phenotype	Type	HPO term (suggested)	Notes / frequency
GI telangiectasias	Physical manifestation	HP:0004389 / related	AR/XLR-predominant

Onset, severity, progression. Mucocutaneous features typically appear in **childhood**; marrow failure often follows in the **first–second decade**; pulmonary and hepatic fibrosis and malignancy are **later** complications. The course is **progressive and multisystem** (Findings F003, F011).

Specific documentation in a genetically confirmed DKCB8 patient:

> "he developed diffuse hyperpigmentation as well as numerous punctate hypopigmented macules, sparse hair, and nail dystrophy, and diagnosis of DC was confirmed with a telomere length assay" — [PMID: 40207375]

Oral involvement extends beyond leukoplakia to lichenoid lesions (reticular, plaque, erosive-ulcerative), and one DC patient developed tongue squamous cell carcinoma at age 25 [PMID: 42625322].

Quality-of-life impact. No DKCB8-specific EQ-5D/SF-36 data exist. By extension from DC/TBD: marrow failure imposes transfusion dependence and infection risk; pulmonary and hepatic fibrosis impair function and survival; malignancy risk requires lifelong surveillance. Impaired reproductive function has been documented in DC (reduced anti-Müllerian hormone, oocyte yield, and fertilization/euploidy rates) [PMID: 32405899].

Section 4 — Genetic / Molecular Information

Causal genes. - ***TYMS*** — thymidylate synthase (HGNC:12441; OMIM 188350), chromosome 18p11.32. Catalyzes dUMP → dTMP. - ***ENOSF1*** — enolase superfamily member 1 (HGNC:24338), the **natural antisense regulator** of *TYMS*, reversely oriented and overlapping (Finding F007).

Pathogenic variants and classification. In DKCB8 the operative lesions are (i) a **loss-of-function *TYMS* coding variant** (the affected 2022 cohort carried heterozygous germline *TYMS* variants) and (ii) rare variants on a **specific *ENOSF1* haplotype** in *trans* [PMID: 35931051]. A distinct DKCB8 case arose from a **structural lesion**: a *TYMS* deletion within a **ring chromosome 18** (partial 18p/18q monosomy) combined with the *ENOSF1* haplotype [PMID: 40207375]. Thus variant classes span **point loss-of-function, structural deletion, and regulatory-haplotype** variation. Formal ACMG/AMP classification of individual DKCB8 alleles is not standardized because the digenic architecture falls outside conventional single-gene rules.

Functional consequence. Net **loss of function** of thymidylate synthase activity. The key mechanistic twist is **epistatic post-transcriptional silencing**: elevated *ENOSF1* suppresses the remaining wild-type *TYMS* allele (Finding F007):

> "post-transcriptional epistatic silencing of TYMS is occurring via elevated ENOSF1" — [PMID: 35931051]
>
> "TYMS expression is regulated by its antisense mRNA, ENOSF1. Disrupted regulation may promote uncontrolled DNA synthesis" — [PMID: 30134598]

Allele frequency. DKCB8-causing configurations are ultra-rare; the *ENOSF1* haplotype-based mechanism means population allele frequencies of individual SNPs do not straightforwardly predict disease. The *TYMS–ENOSF1* region is well studied pharmacogenomically (e.g., 5′-UTR VNTR; rs495139; rs3819102) [PMID: 22496803; 30134598; 35631247].

Somatic vs germline. The DKCB8 lesions are **germline**. (Somatic clonal hematopoiesis can arise secondarily in DC marrow — see Sections 6 and 11 [PMID: 32736377].)

Modifier genes / epigenetics. *ENOSF1* itself functions as the principal modifier via antisense regulation. No additional DKCB8-specific modifier genes or DNA-methylation signatures have been reported.

Chromosomal abnormalities. A ring chromosome 18 with partial 18p/18q monosomy encompassing *TYMS* has been reported as one route to DKCB8 [PMID: 40207375].

Section 5 — Environmental Information

Environmental factors. No toxin, radiation, or occupational exposure is required for DKCB8. The clinically relevant environmental interaction is **pharmacologic**: fluoropyrimidines (5-FU, capecitabine, FdUMP metabolites) and antifolates (raltitrexed, methotrexate) inhibit thymidylate synthase and would be expected to compound the pre-existing deficiency (Finding F009) [PMID: 35931051; 27569869; 15930305; 25245820].

Lifestyle factors. Dietary folate / one-carbon metabolism is mechanistically linked because TYMS uses 5,10-methylenetetrahydrofolate; folate status modulates thymidylate biosynthesis and genome integrity in model systems [PMID: 37183313; 28461497]. Whether folate supplementation modifies DKCB8 severity is untested. As with all DC/TBD, smoking is a general risk factor for squamous carcinogenesis and should be avoided.

Infectious agents. Not applicable — DKCB8 is a genetic disorder with no infectious etiology. (Recurrent infections occur secondary to marrow failure/immune dysfunction.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Germline digenic lesion** — a loss-of-function *TYMS* coding (or structural-deletion) allele is inherited from one parent, **and** a specific *ENOSF1* antisense haplotype with rare variants is inherited in *trans* from the other parent → **results in** a genotype in which only one functional *TYMS* allele would otherwise remain. *(demonstrated)* [PMID: 35931051]
2. Elevated *ENOSF1* antisense activity **leads to** post-transcriptional silencing of the remaining wild-type *TYMS* allele → **results in** severe thymidylate synthase deficiency. *(demonstrated*

by gene-rescue) [PMID: 35931051;

P 30134598]

3. Thymidylate synthase deficiency **leads to** failure of *de novo* dUMP → dTMP conversion → **results in** dTMP/dTTP depletion and an imbalanced (altered) cellular dNTP pool. *(demonstrated)* [PMID: 35931051]
4. dTMP depletion (relative dUTP excess) **leads to uracil misincorporation into DNA** → which is excised by **base-excision repair (BER)** → **generates strand-break intermediates**. *(supported by mechanistic literature)* [PMID: 18773878]
5. BER intermediates and thymineless stress **lead to** replication-fork stalling/collapse and DNA double-strand breaks, activating homologous recombination (RAD51), RPA2 and γ-H2AX → **results in** genotoxic replication stress. *(inferred from thymidylate-stress models)* [PMID: 18773878;

P 25245820 ;

P 2839770 ;

P 15930305]

> "Thymidylate deprivation increases dUTP and uracil in DNA, which is removed by base excision repair (BER)" — [PMID: 18773878]

6. Genotoxic stress and defective transcription **lead to abnormal telomere maintenance** → **results in** accelerated telomere attrition and replicative senescence in high-turnover stem/progenitor compartments. *(demonstrated)* [PMID: 35931051]

> "These defects in the nucleotide metabolism pathway resulted in genotoxic stress, defective transcription, and abnormal telomere maintenance" — [PMID: 35931051]

7. Stem-cell attrition in bone marrow and epithelia **leads to** the clinical phenotype: **bone marrow failure** and the **mucocutaneous triad**; over time → **pulmonary/hepatic fibrosis**. *(demonstrated clinically)* [PMID: 34852175;

P 31754622]

8. **Branch:** chronic genotoxic stress and stem-cell depletion **lead to** selective pressure favoring **clonal hematopoiesis**, and to **squamous and myeloid malignancy** (MDS, AML, head/neck and anogenital SCC). *(demonstrated in DC/TBD populations)* [PMID: 32736377; P 36091172]

Categorical detail

- **Molecular pathways:** nucleotide/pyrimidine (thymidylate) biosynthesis; one-carbon/folate metabolism (5,10-methylene-THF cofactor); DNA base-excision repair and homologous recombination; telomere maintenance. **KEGG:** pyrimidine metabolism (hsa00240); one-carbon pool by folate (hsa00670).
- **Cellular processes (GO suggestions):** GO:0006231 (dTMP biosynthetic process); GO:0006281 (DNA repair); GO:0006284 (base-excision repair); GO:0000723 (telomere maintenance); GO:0090399 (replicative senescence); GO:0006974 (DNA-damage response).
- **Protein dysfunction:** loss of thymidylate synthase catalytic output; the enzyme normally functions as a homodimer using 5,10-methylene-THF (UniProt P04818).
- **Metabolic changes:** dTMP/dTTP depletion, dUMP/dUTP accumulation, dNTP-pool imbalance [PMID: 35931051].
- **Tissue-damage mechanism:** thymineless/genotoxic replication stress with DNA strand breaks (γ-H2AX, RPA2) [PMID: 18773878; P 25245820].
- **Immune involvement:** secondary immunodeficiency from marrow failure; broader DC literature notes T-cell/immune-surveillance defects relevant to cancer risk [PMID: 41868676].

Cell types (CL suggestions): hematopoietic stem cell (CL:0000037), common myeloid progenitor (CL:0000049), keratinocyte (CL:0000312), oral mucosal epithelial cell, hepatic stellate cell (CL:0000632, fibrosis), type II pneumocyte (CL:0002063).

Section 7 — Anatomical Structures Affected

Organ level. - *Primary:* bone marrow / hematopoietic system (UBERON:0002371, bone marrow), skin (UBERON:0002097), nails, oral mucosa (UBERON:0003729). - *Secondary:* lungs (pulmonary fibrosis, AVMs; UBERON:0002048), liver (fibrosis; UBERON:0002107), gastrointestinal tract (telangiectasias; UBERON:0001555), eyes (strabismus; UBERON:0000970). - *Body systems:* hematopoietic/immune, integumentary, respiratory, digestive/hepatobiliary.

Tissue and cell level. Predominantly **rapidly proliferating epithelial and hematopoietic tissues**. Affected cell populations (CL): hematopoietic stem/progenitor cells (CL:0000037), keratinocytes (CL:0000312), oral mucosal epithelium, hepatic stellate cells and pulmonary fibroblasts (fibrotic remodeling), type II pneumocytes (CL:0002063).

Subcellular level (GO Cellular Component). Nucleus (GO:0005634) and cytoplasm (GO:0005737, site of thymidylate synthase and dNTP synthesis); telomeric chromosomal ends (GO:0000781, chromosome, telomeric region).

Localization / lateralization. Skin pigmentation is typically diffuse/reticulate; oral leukoplakia and lichenoid lesions affect the tongue and buccal mucosa; pulmonary and hepatic fibrosis are **bilateral/diffuse**. No consistent lateralization.

Section 8 — Temporal Development

Onset. Congenital predisposition with **childhood-onset** clinical manifestations. Mucocutaneous features usually appear first (childhood), followed by marrow failure (Finding F011). The reported DKCB8 ring-18 case presented in early childhood [PMID: 40207375].

Progression. Chronic, progressive, multisystem. Typical trajectory: mucocutaneous triad (childhood) → bone marrow failure (first–second decade) → pulmonary/hepatic fibrosis and malignancy (later). Age-at-diagnosis data for DC/TBD: median **19.4 years (range 0–71.6)** in the NCI cohort [PMID: 34852175] and **9 years** in the Canadian pediatric registry [PMID: 42267950].

> "median age at diagnosis 19.4 years [range 0 to 71.6]" — [PMID: 34852175]

Patterns. No spontaneous remission. Marrow failure can be transiently stabilized by androgens and cured (hematologically) by HSCT. **Critical windows:** early recognition of marrow failure to time HSCT before severe non-hematopoietic organ damage accrues; avoidance of TYMS-inhibiting chemotherapy at all times.

Section 9 — Inheritance and Population

Epidemiology. DKCB8 is **ultra-rare**: originally 8 independent families [PMID: 35931051] plus rare subsequent case reports [PMID: 40207375]. Dyskeratosis congenita overall is estimated at **~1 per million** (Finding F011).

Inheritance. Autosomal recessive but digenic — a *TYMS* coding LOF allele plus a *trans* *ENOSF1* antisense haplotype. Classic single-gene recessive segregation is **not** observed; one parent carries a wild-type *TYMS* coding sequence [PMID: 35931051].

> "In a cohort of eight independent DC-affected families, we have identified a remarkable series of heterozygous germline variants in the gene encoding thymidylate synthase (TYMS)" — [PMID: 35931051]

Penetrance / expressivity. Presumed high penetrance when both genetic requirements are met; expressivity is variable, consistent with the broader DC/TBD spectrum. Formal penetrance estimates are unavailable given the small case count.

Anticipation / mosaicism / founder effects. Not established for DKCB8. (Anticipation is a general feature of telomere biology disorders due to progressive telomere shortening across generations but has not been specifically quantified for DKCB8.)

Consanguinity / carrier frequency. Not specifically reported for DKCB8; the digenic mechanism complicates carrier-frequency estimation. The permissive *ENOSF1* haplotype is common in the population, whereas the *TYMS* LOF allele is rare — so disease requires the specific *trans* combination.

Population demographics. No defined ethnic predilection reported. Sex ratio not established. Within DC/TBD broadly, autosomal-recessive/X-linked forms tend to present earlier and more severely (relevant to DKCB8) [PMID: 34852175].

Section 10 — Diagnostics

Diagnostic strategy (Finding F006). Two-step: (1) **telomere length screening**, then (2) **molecular confirmation**.

Telomere length. Measured by **flow-FISH**; very short telomeres (**<1st percentile** of age-matched controls, or age-modified thresholds such as <6.5 kb in patients >40 y) are the key screening biomarker.

> "TL was considered suspicious once below the 10th percentile of normal individuals (standard screening) or if below 6.5 kb in patients >40 years (extended screening). In cases with shortened TL, next generation sequencing (NGS) for TBD-associated genes was performed" — [PMID: 37096215]

For the confirmed DKCB8 case, a telomere length assay confirmed DC and the *TYMS* deletion was identified genetically [PMID: 40207375].

Genetic testing. WES/NGS with **segregation analysis** is standard for DC/TBD [PMID: 42557666]. Because DKCB8 is **digenic** (*TYMS* coding + *ENOSF1* antisense haplotype in *trans*), **standard single-gene panels may miss it — targeted genomic sequencing of the *TYMS*–*ENOSF1* locus** is required.

> "Targeted genomic sequencing identified a specific haplotype and rare variants in the naturally occurring TYMS antisense regulator ENOSF1" — [PMID: 35931051]

Chromosomal microarray / karyotyping is warranted when a structural lesion (e.g., ring chromosome 18) is suspected [PMID: 40207375].

Ancillary testing / organ surveillance. - **Liver:** transient elastography detects subclinical fibrosis in ~**88.8%** of TBD patients [PMID: 34565437]. - **Lung:** restrictive spirometry and reduced DLCO in **42%** of DC patients [PMID: 31754622]. - **Marrow:** CBC, bone marrow aspirate/biopsy for cytopenias and MDS surveillance. - **In vitro corroboration:** patient lymphoblastoid cells show TYMS deficiency, altered dNTP pools, and **5-FU hypersensitivity** [PMID: 35931051].

Clinical criteria / differential diagnosis. Diagnosis rests on the mucocutaneous triad + marrow failure + very short telomeres + molecular confirmation. Differential: other DC genotypes (DKC1, TERT, TERC, RTEL1, TINF2, PARN), Fanconi anemia, Shwachman–Diamond syndrome, Hoyeraal–Hreidarsson syndrome, and acquired aplastic anemia [PMID: 35605178; 36091172; 37507252].

Screening. Cascade telomere-length + targeted molecular testing of at-risk relatives; no newborn screening exists for this ultra-rare entity [PMID: 36286734].

Section 11 — Outcome / Prognosis

Survival / mortality (DC/TBD context; Finding F005). In 231 DC/TBD individuals (NCI IBMFS study): **42% deceased**, median overall survival **52.8 years (95% CI 45.5–57.6)**; transplant-free median survival **45.3 years (95% CI 37.4–52.1)**.

> "42% of patients were deceased with a median overall survival (OS) of 52.8 years (95% confidence interval [CI] 45.5-57.6)" — [PMID: 34852175]

AR/XLR forms (relevant to DKCB8) carry the worst prognosis:

> "Severe bone marrow failure (BMF), severe liver disease, and gastrointestinal telangiectasias were more prevalent in AR/XLR or TINF2 disease... After adjusting for age at DC/TBD diagnosis, we observed the highest cancer risk in AR/XLR individuals" — [PMID: 34852175]

Cancer risk. Increased risk of **MDS, AML, and solid tumors** — especially **head/neck and anogenital squamous cell carcinoma**; clonal hematopoiesis contributes to leukemia risk [PMID: 36091172; 32736377].

Morbidity / disease course. Complications include marrow failure, pulmonary fibrosis, pulmonary AVMs, liver fibrosis, hepatopulmonary syndrome, and GI telangiectasias [PMID: 40356079; 31754622;

].

P 34565437

Even after curative HSCT for marrow failure, **late mortality from pulmonary and hepatic fibrosis remains high** (e.g., 4/7 died at median 10 years post-HSCT) [PMID: 40356079].

Prognostic factors. Younger age at diagnosis and severe BMF predict worse outcome (pediatric registry: severe BMF associated HR 7.5 for mortality) [PMID: 42267950]. Baseline PFT abnormalities predict pulmonary outcomes [PMID: 31754622].

Treatment-related prognostic note. Androgen therapy improves cytopenias but creates an **atherogenic lipoprotein profile** (↓HDL-C, HDL particle number/size; ↑LDL-C, apoB; all $p<0.001$), warranting cardiovascular monitoring [PMID: 34929494].

Section 12 — Treatment

No disease-specific/curative therapy exists for the underlying TYMS deficiency; management follows DC/TBD principles (Finding F004).

Pharmacotherapy — androgens (first-line for cytopenias). Danazol/oxymetholone can improve hematologic parameters (NCIT: androgen therapy C1516; danazol C494; oxymetholone C716).

> "Although hematological defects can respond to danazol/oxymetholone, the only current curative treatment for these is hematopoietic stem cell transplantation (HSCT) using fludarabine-based conditioning protocols" — [PMID: 35929966]

Monitor lipids/cardiovascular risk on androgens [PMID: 34929494].

Hematopoietic stem cell transplantation (only cure for marrow failure). Fludarabine-based reduced-intensity conditioning (RIC) — often with alemtuzumab, minimizing/avoiding radiation and alkylators — is standard because of mucosal, vascular, pulmonary, and hepatic fragility (NCIT: hematopoietic stem cell transplantation C15431; fludarabine C1094).

> "Because of toxicity after myeloablative conditioning, RIC is becoming standard for HCT in DKC. These results suggest that RIC regimen is feasible and safe for patients with DKC and does not accelerate pulmonary damage in the short-to-medium term after HCT" — [PMID: 34086408]

A prospective single-arm trial found **TBI is dispensable** for DC/TBD-associated marrow failure [PMID: 39002862]. HSCT **does not correct non-hematopoietic complications**, and late pulmonary/hepatic fibrosis remains the leading cause of post-HSCT mortality [PMID: 40356079].

Pharmacogenomics — a DKCB8-specific caution. Because thymidylate synthase is the target of **fluoropyrimidines (5-FU, capecitabine)** and antifolates, and DKCB8 cells are **hypersensitive to 5-FU**, these agents should be **avoided or used with extreme caution** (Finding F009):

> "hypersensitivity to the TYMS-specific inhibitor 5-fluorouracil" — [PMID: 35931051]
>
> "variants in genes of the 5-FU metabolic pathway, including TYMS, MTHFR and DPYD also influenced capecitabine efficacy and toxicity" — [PMID: 27569869]

This is a mechanistically-inferred recommendation of high clinical relevance, especially given the elevated malignancy risk in DC (where fluoropyrimidines might otherwise be considered).

Supportive/experimental. Transfusion support, infection prophylaxis, malignancy surveillance, organ-specific management (pulmonary, hepatic). Emerging DC therapeutics under study include PAPD5 inhibitors and other telomere-directed agents (broader DC pipeline) [PMID: 35605178]. Lung transplantation may be considered for pulmonary failure [PMID: 28407835]. No DKCB8-specific gene, cell, or RNA therapy exists.

Section 13 — Prevention

- **Primary prevention:** Not applicable to disease occurrence (genetic). **Genetic counseling** for at-risk families is the principal tool; the digenic mechanism complicates conventional recurrence-risk counseling and requires locus-specific interpretation [PMID: 36286734].

- **Secondary prevention: Telomere-length screening (flow-FISH) + targeted molecular testing** for cascade identification of affected/at-risk relatives; early detection of subclinical liver and lung fibrosis (elastography, PFTs) [PMID: 37096215; 34565437; 31754622].
- **Tertiary prevention:** Malignancy surveillance (skin, oral, anogenital, marrow), androgen therapy for cytopenias, timely HSCT, cardiovascular monitoring on androgens, and — critically — **avoidance of fluoropyrimidine/antifolate chemotherapy** [PMID: 35929966; 34929494; 35931051].
- **Behavioral:** Smoking cessation (squamous cancer risk); consideration of folate/one-carbon status (mechanistically linked, unproven for DKCB8).
- **Reproductive:** Fertility preservation counseling given documented reproductive impairment in DC [PMID: 32405899]; prenatal/preimplantation testing is theoretically possible once familial variants are defined.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs (Finding F010):** *TYMS* is an **essential, ancient enzyme** conserved from bacteria and yeast to humans. Orthologs include mouse *Tyms* (NCBI Gene 22171), zebrafish *tyms*, *Drosophila*, and yeast (*CDC21/TMP1*).
- **Natural disease: No naturally occurring animal disease** equivalent to DKCB8 has been reported in OMIA. The digenic *TYMS–ENOSF1* antisense architecture appears to be a human-specific configuration.
- **Comparative biology:** Thymidylate-stress phenotypes (chromosomal DNA fragmentation, thymineless death) are conserved and well documented in mouse cells and *Drosophila*, supporting the mechanistic model [PMID: 2839770; 37183313].
- **Zoonotic potential:** Not applicable (non-infectious genetic disorder).

Section 15 — Model Organisms

Primary model (Finding F010): patient-derived lymphoblastoid cell lines (LCLs) — the established DKCB8 experimental system:

> "Lymphoblastoid cells from affected probands have severe TYMS deficiency, altered cellular deoxyribonucleotide triphosphate pools, and hypersensitivity to the TYMS-specific inhibitor 5-fluorouracil" — [PMID: 35931051]

These cells recapitulate TYMS deficiency, dNTP-pool imbalance, 5-FU hypersensitivity, genotoxic stress, and abnormal telomere maintenance, and gene-rescue experiments confirmed **ENOSF1-mediated silencing** of TYMS.

Related/supporting models. - **Chemically induced thymidylate stress:** mouse FM3A cells and 5-FdUrd/raltitrexed models reproduce thymineless DNA damage [PMID: 2839770;

P 15930305

;

P 25245820

].

- ***Drosophila*** gene–nutrient (vitamin B6 / SHMT / TS) models link one-carbon metabolism to genome integrity [PMID: 37183313]. - **Broader DC telomere biology** has been modeled in mice and zebrafish for other DC genes (e.g., *dkc1*), but **no TYMS-specific animal model** of DKCB8 exists.

Limitations. No animal model captures the human-specific digenic *TYMS–ENOSF1* antisense mechanism; LCLs do not reproduce tissue-level fibrosis or malignancy. Development of a humanized *TYMS–ENOSF1* model is an open need.

Mechanistic Model (synthesis)

```
   [TYMS LOF coding/deletion allele]        [ENOSF1 permissive haplotype + rare variants]
              (parent A)                                     (parent B, in trans)
                     \                                         /
                      \____________ DIGENIC GENOTYPE _________/
                                          |
                         Elevated ENOSF1 antisense activity
                                          |
                         Post-transcriptional silencing of WT TYMS
                                          |
                         SEVERE THYMIDYLATE SYNTHASE DEFICIENCY
                                          |
                 dUMP up / dTMP down , imbalanced dNTP pool
                                          |
                 Uracil misincorporation into DNA  ->  BER strand breaks
                                          |
                 Replication-fork collapse, gamma-H2AX/RPA2, HR (RAD51)
                                          |
                 GENOTOXIC STRESS + defective transcription
                                          |
                 ABNORMAL TELOMERE MAINTENANCE -> stem-cell attrition
                        /                 |                 \
          Bone marrow failure   Mucocutaneous triad   Pulmonary/hepatic fibrosis
                        \                                   /
                          Clonal hematopoiesis -> MDS/AML
                          Epithelial dysplasia -> SCC (head/neck, anogenital)
```

Upstream drivers are the **digenic lesion → TYMS deficiency → nucleotide imbalance** ; downstream effects are **telomere dysfunction → multisystem stem-cell failure and malignancy**. The pharmacogenomic branch (fluoropyrimidine/antifolate hypersensitivity) intersects the upstream node directly.

Evidence Base

PMID	Study (abbrev.)	Role in this report
35931051	*Germline thymidylate synthase deficiency... causes DC*	**Foundational** — digenic TYMS–ENOSF1 mechanism, LCL model, 5-FU hypersensitivity, telomere defect (F001, F002, F007, F008, F009, F010, F011)
40207375	*TYMS-ENOSF1 DC in ring chromosome 18*	Independent confirmation; structural-deletion route; phenotype (F001, F003)
18773878	*DNA damage/HR from thymidylate deprivation*	Uracil/BER/strand-break step (F002)
25245820	*Raltitrexed/TS inhibition, DNA damage*	Thymidylate-stress DNA-damage mechanism (F002)
30134598	*rs495139 in TYMS-ENOSF1 region*	ENOSF1 antisense regulation of TYMS (F007)
42625322	*Oral lichenoid lesions in DC*	Mucocutaneous triad; oral spectrum; tongue SCC (F003)
35929966	*Biology and management of DC*	Androgens + fludarabine-based HSCT only cure (F004)
34086408	*RIC-based HCT for DC*	RIC standard; feasible/safe (F004)
39002862	*RIC without radiation trial*	TBI dispensable for DC/TBD BMF (F004)
40356079	*Late complications post-HSCT in DC*	High late mortality from PF/LF (F004, F005)
34852175	*Disease progression/ outcomes in TBD*	AR/XLR worst prognosis; survival; age at diagnosis (F005, F011)
36091172	*FA and DC/TBD genomic instability*	Cancer spectrum (F005)
32736377	*Clonal hematopoiesis in IBMFS*	Clonal hematopoiesis → leukemia risk (F005)
34565437	*Transient elastography in cryptic DC*	Subclinical liver fibrosis 88.8% (F006)
31754622	*PFTs in DC*	Restrictive/DLCO abnormalities 42% (F006)
37096215		TL-first, then-NGS diagnostic strategy (F006)

PMID	Study (abbrev.)	Role in this report
	Telomere length screening, age-modified	
42557666	*TERT-associated DC characterization*	WES + segregation + flow-FISH paradigm (F006)
27569869	*Pharmacogenetics of capecitabine*	TYMS as fluoropyrimidine PGx determinant (F009)
22496803	*TYMS genetic region polymorphisms*	TYMS pharmacogenetics (F009)
34929494	*Lipoprotein alterations from androgens in DC*	Androgen cardiovascular risk (F005)
42267950	*Canadian Inherited Marrow Failure Registry — DC*	Pediatric outcomes; median dx age 9 y; severe BMF HR 7.5 (F011)
37183313	*B6/SHMT gene-nutrient interaction (Drosophila)*	One-carbon/folate link (F009)
2839770	*Chromosomal DNA degradation from thymidylate stress*	Conserved thymineless-death model (F002)

Limitations and Knowledge Gaps

1. **Very small evidence base for DKCB8 specifically.** Almost all molecular evidence derives from a single landmark study (8 families) [PMID: 35931051] plus one case report [PMID: 40207375]. Much of the clinical, prognostic, and treatment detail is **extrapolated from the broader DC/TBD population** rather than measured in DKCB8 patients.
2. **No DKCB8-specific epidemiology.** Prevalence, incidence, sex ratio, penetrance, and ethnic distribution are unknown; only DC-wide estimates (~1/million) are available.
3. **Digenic classification challenges.** ACMG/AMP frameworks are built for single-gene disease; the *TYMS* coding + *ENOSF1* haplotype architecture is not readily scored, and the permissive *ENOSF1* haplotype's precise functional variants remain incompletely defined.

4. **No animal model** captures the human-specific antisense mechanism; tissue-level pathology (fibrosis, cancer) cannot be studied in the LCL system.
5. **Pharmacogenomic caution is inferred**, not clinically demonstrated in DKCB8 patients — based on in vitro 5-FU hypersensitivity and TYMS biology.
6. **Folate/one-carbon modulation** as a potential modifier is biologically plausible but untested in DKCB8.
7. **Long-term natural history and treatment response specific to DKCB8** (HSCT outcomes, cancer incidence) are unknown due to case scarcity.

Proposed Follow-up Experiments / Actions

1. **Establish a DKCB8 patient registry / GeneMatcher effort** to aggregate cases, define natural history, penetrance, and genotype–phenotype correlations across *TYMS* LOF and *ENOSF1* haplotype configurations.
2. **Functional dissection of the permissive *ENOSF1* haplotype** — CRISPR/allele-specific editing in isogenic LCLs or iPSCs to identify the causal regulatory variants and quantify their effect on TYMS silencing.
3. **Generate a humanized *TYMS–ENOSF1* model** (iPSC-derived hematopoietic/epithelial organoids or a humanized-locus mouse) to recapitulate telomere attrition, marrow failure, and fibrosis.
4. **Systematic telomere-length + dNTP-pool profiling** across tissues to test whether nucleotide imbalance precedes telomere shortening (order-of-events causality).
5. **Prospective pharmacovigilance / contraindication guidance:** formalize avoidance of fluoropyrimidines and antifolates in DKCB8 clinical protocols; test whether thymidine/dTMP supplementation rescues patient-cell phenotypes as a candidate therapeutic strategy.
6. **Folate/one-carbon intervention studies** in patient cells to determine whether cofactor availability modulates residual TYMS activity and genotoxic stress.
7. **Incorporate *TYMS–ENOSF1* locus-targeted sequencing** into standard DC/TBD diagnostic pipelines so digenic cases are not missed by conventional single-gene panels.

Report compiled from 11 confirmed findings across 5 investigation iterations and 44 reviewed papers. Evidence source types are indicated inline: human clinical (patient cohorts/case reports), in vitro (patient LCLs), model organism (mouse/Drosophila thymidylate-stress systems), and computational/database (MONDO, OMIM cross-references).
